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Lab #4:

#Stack using arrays :

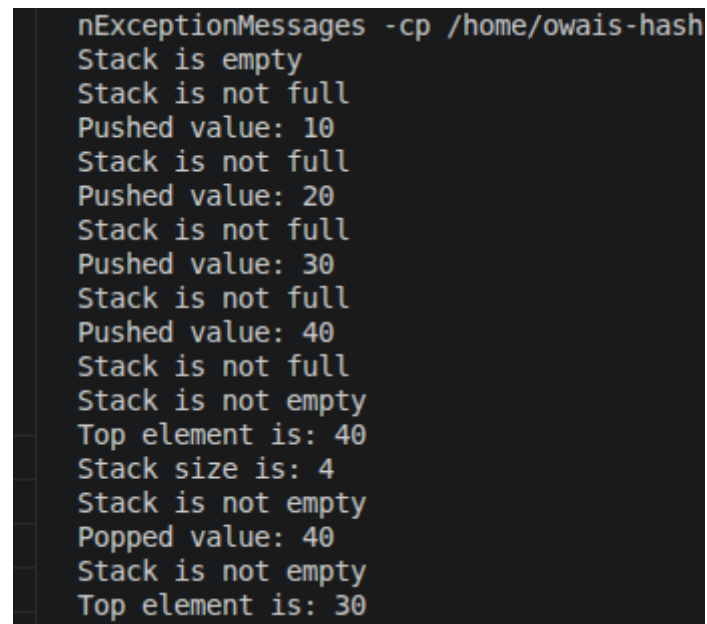
```
public class stack {
    private static int maxSize;
    int [] stackArray = new int[maxSize];
    int top = -1;
    public stack(int size){
        this.maxSize = size;
    }
    public boolean isEmpty(){
        if(top== -1){
            System.out.println("Stack is empty");
            return true;
        }
        else{
            System.out.println("Stack is not empty");
            return false;
        }
    }
    public boolean isFull(){
        if(top==maxSize-1){
            System.out.println("Stack is full");
            return true;
        }
        else{
            System.out.println("Stack is not full");
            return false;
        }
    }
    public void push(int value){
        if(isFull()){
            System.out.println("Stack is full, cannot push value");
        }
        else{
            top++;
            stackArray[top]=value;
            System.out.println("Pushed value: "+value);
        }
    }
    public int pop(){
        if(isEmpty()){
            System.out.println("Stack is empty, cannot pop value");
            return -1;
        }
        else{
            int value = stackArray[top];
            top--;
            System.out.println("Popped value: "+value);
            return value;
        }
    }
    public int peek(){
        if(isEmpty()){
            System.out.println("Stack is empty, cannot peek value");
            return -1;
        }
        else{
            System.out.println("Top element is: "+stackArray[top]);
            return stackArray[top];
        }
    }
    public int size(){
        return top+1;
    }
}
```

```

public static void main(String[] args) {
    stack s = new stack(5);
    s.isEmpty();
    s.push(10);
    s.push(20);
    s.push(30);
    s.push(40);
    s.isFull();
    s.peek();
    System.out.println("Stack size is: "+s.size());
    s.pop();
    s.peek();}}

```

OUTPUT :



```

nExceptionMessages -cp /home/owais-hash
Stack is empty
Stack is not full
Pushed value: 10
Stack is not full
Pushed value: 20
Stack is not full
Pushed value: 30
Stack is not full
Pushed value: 40
Stack is not full
Stack is not empty
Top element is: 40
Stack size is: 4
Stack is not empty
Popped value: 40
Stack is not empty
Top element is: 30

```

Implement two stacks using one array :

```

public class two_stack {
    int maxSize;
    int [] stackArray ;
    int top1;
    int top2;
    public two_stack(int size){
        this.maxSize = size;
        stackArray = new int[maxSize];
        top1=-1;
        top2=maxSize;}
    public boolean isEmpty1(){
        if(top1== -1){
            System.out.println("Stack 1 is empty");
            return true;}
        else{
            System.out.println("Stack 1 is not empty");
            return false;}}

```

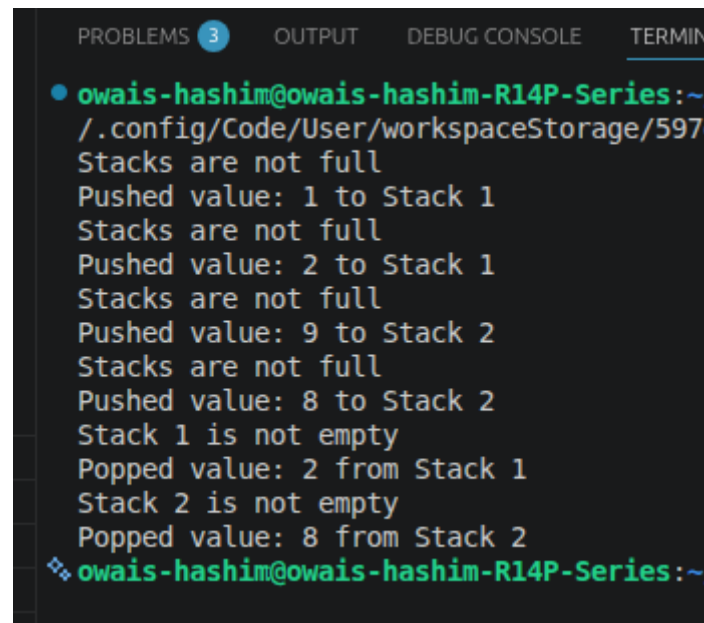
```

public boolean isEmpty2(){
    if(top2==maxSize){
        System.out.println("Stack 2 is empty");
        return true;}
    else{
        System.out.println("Stack 2 is not empty");
        return false;}}
public boolean isFull(){
    if(top1+1==top2){
        System.out.println("Stacks are full");
        return true;}
    else{
        System.out.println("Stacks are not full");
        return false;}}
public void push1(int value){
    if(isFull()){
        System.out.println("Stacks are full, cannot push value to Stack 1");}
    else{
        top1++;
        stackArray[top1]=value;
        System.out.println("Pushed value: "+value+" to Stack 1");}}
public void push2(int value){
    if(isFull()){
        System.out.println("Stacks are full, cannot push value to Stack 2");}
    else{
        top2--;
        stackArray[top2]=value;
        System.out.println("Pushed value: "+value+" to Stack 2");}}
public int pop1(){
    if(isEmpty1()){
        System.out.println("Stack 1 is empty, cannot pop value");
        return -1;}
    else{
        int value = stackArray[top1];
        top1--;
        System.out.println("Popped value: "+value+" from Stack 1");
        return value;}}
public int pop2(){
    if(isEmpty2()){
        System.out.println("Stack 2 is empty, cannot pop value");
        return -1;}
    else{
        int value = stackArray[top2];
        top2++;
        System.out.println("Popped value: "+value+" from Stack 2");
        return value;}}
public static void main(String[] args) {
    two_stack ts = new two_stack(10);
    ts.push1(1);
    ts.push1(2);
    ts.push2(9);
    ts.push2(8);
}

```

```
ts.pop1();
ts.pop2();}}
```

OUTPUT :



```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL
• owais-hashim@owais-hashim-R14P-Series:~
/.config/Code/User/workspaceStorage/597
Stacks are not full
Pushed value: 1 to Stack 1
Stacks are not full
Pushed value: 2 to Stack 1
Stacks are not full
Pushed value: 9 to Stack 2
Stacks are not full
Pushed value: 8 to Stack 2
Stack 1 is not empty
Popped value: 2 from Stack 1
Stack 2 is not empty
Popped value: 8 from Stack 2
❖ owais-hashim@owais-hashim-R14P-Series:~
```

#Stack using Linked List :

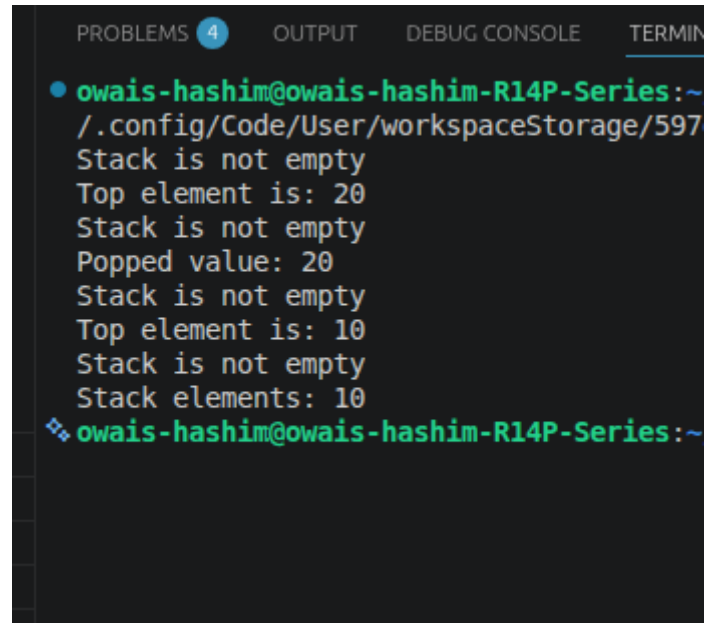
```
class Node {
private int data;
private Node next;
public Node() {
data = 0;
next = null;}
public Node(int val , Node n) {
data = val;
next = n;}
public void setdata(int val){
data = val;}
public int getdata() {
return data;}
public Node getNext() {
return next;}
public void setnext(Node n) {
next = n;}}
public class linked_stack {
protected Node top;
protected int size;
public linked_stack(){
top = null;
size = 0; }
public boolean isEmpty(){
if(top == null){
System.out.println("Stack is empty");
return true;}
else{
```

```

System.out.println("Stack is not empty");
return false;}}
public int getSize(){
return size;}
public void push(int value){
Node newNode = new Node(value, top);
if(top == null){
top = newNode;
} else {
newNode.setnext(top);
top = newNode;}
size++;}
public int pop(){
if(isEmpty()){
System.out.println("Stack is empty, cannot pop value");
return -1;}
else{
int value = top.getdata();
top = top.getnext();
size--;
System.out.println("Popped value: "+value);
return value;}}
public int peek(){
if(isEmpty()){
System.out.println("Stack is empty, cannot peek value");
return -1;}
else{
System.out.println("Top element is: "+top.getdata());
return top.getdata();}}
public void display(){
if(isEmpty()){
System.out.println("Stack is empty, cannot display values");}
else{
Node current = top;
System.out.print("Stack elements: ");
while(current != null){
System.out.print(current.getdata() + " ");
current = current.getnext();}
System.out.println();}}
public static void main(String[] args) {
linked_stack ls = new linked_stack();
ls.push(10);
ls.push(20);
ls.peek();
ls.pop();
ls.peek();
ls.display();}}

```

OUTPUT :



A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS' (with a blue circle containing the number 4), 'OUTPUT', 'DEBUG CONSOLE', and 'TERMINAL'. The terminal shows a series of commands and their outputs. The prompt is 'owais-hashim@owais-hashim-R14P-Series:~'. The first command is './.config/Code/User/workspaceStorage/597', which produces the following output: 'Stack is not empty', 'Top element is: 20', 'Stack is not empty', 'Popped value: 20', 'Stack is not empty', 'Top element is: 10', 'Stack is not empty', and 'Stack elements: 10'. The second command is a diamond-shaped icon followed by the prompt, which produces no visible output.

```
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL
• owais-hashim@owais-hashim-R14P-Series:~
  /.config/Code/User/workspaceStorage/597
  Stack is not empty
  Top element is: 20
  Stack is not empty
  Popped value: 20
  Stack is not empty
  Top element is: 10
  Stack is not empty
  Stack elements: 10
❖ owais-hashim@owais-hashim-R14P-Series:~
```